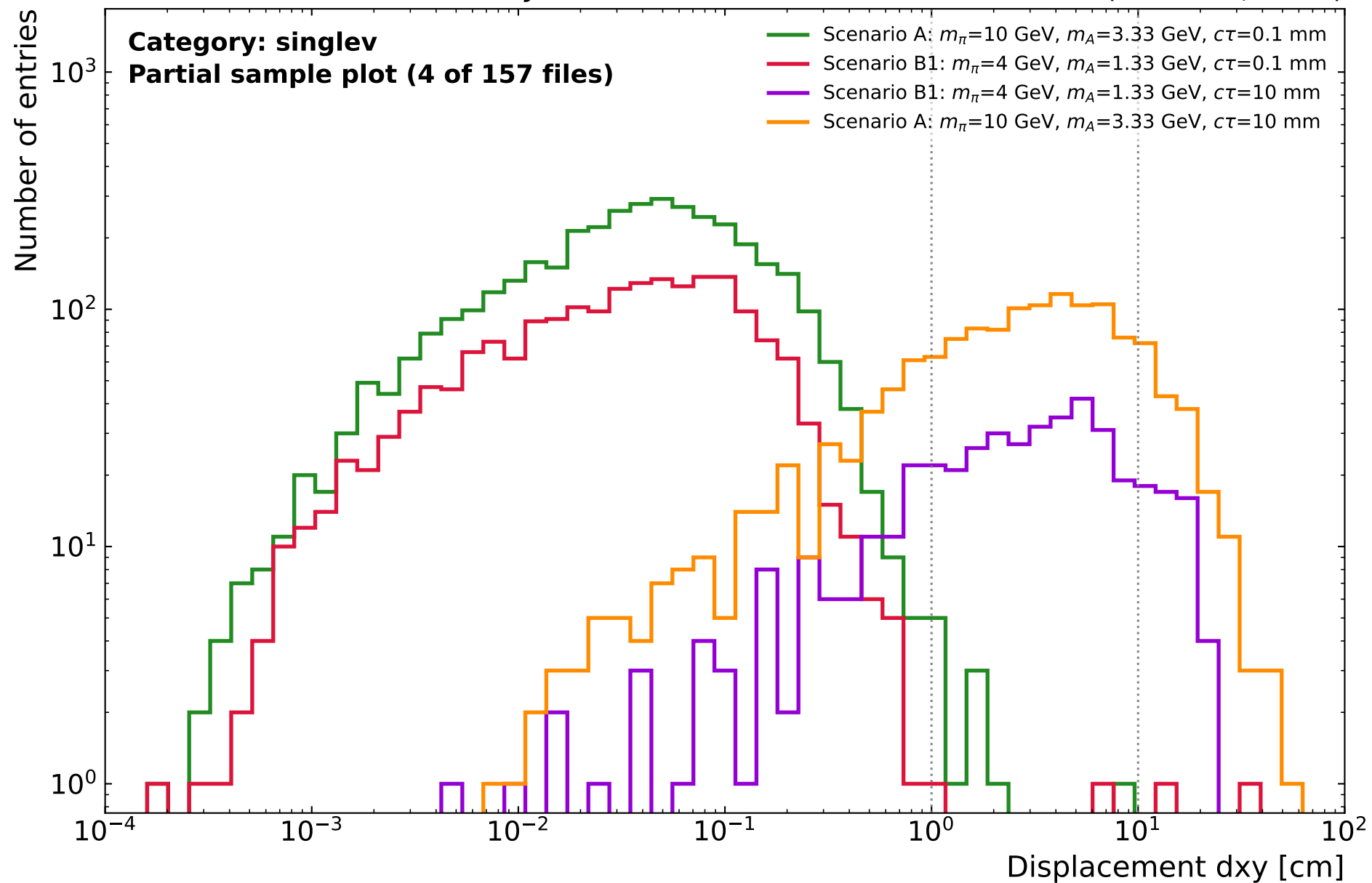


singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
dotted: dxy category edges (1, 10 cm)

CMS *Simulation Preliminary*

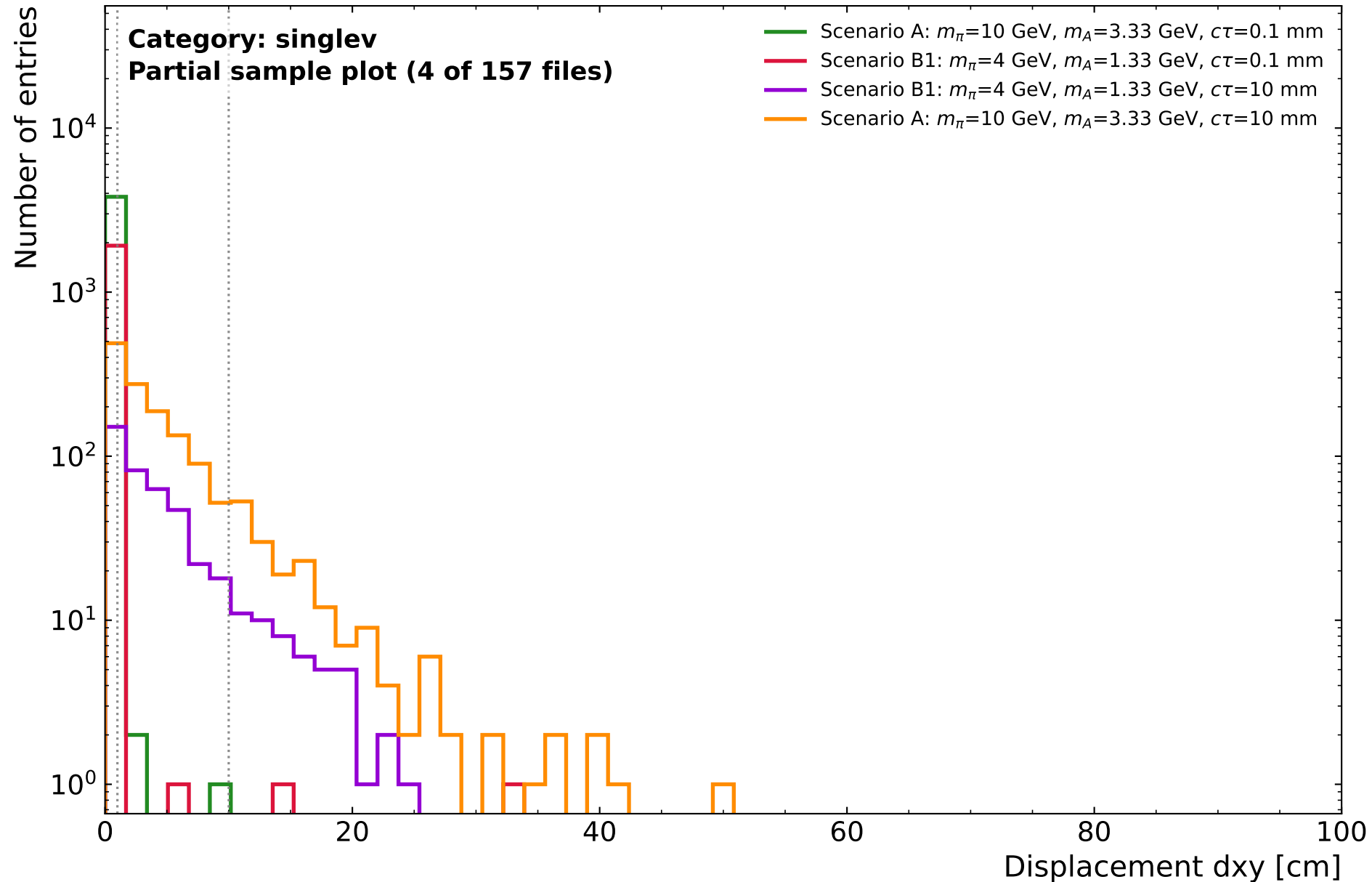
(13.6 TeV, 2024)



singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
dotted: dxy category edges (1, 10 cm)

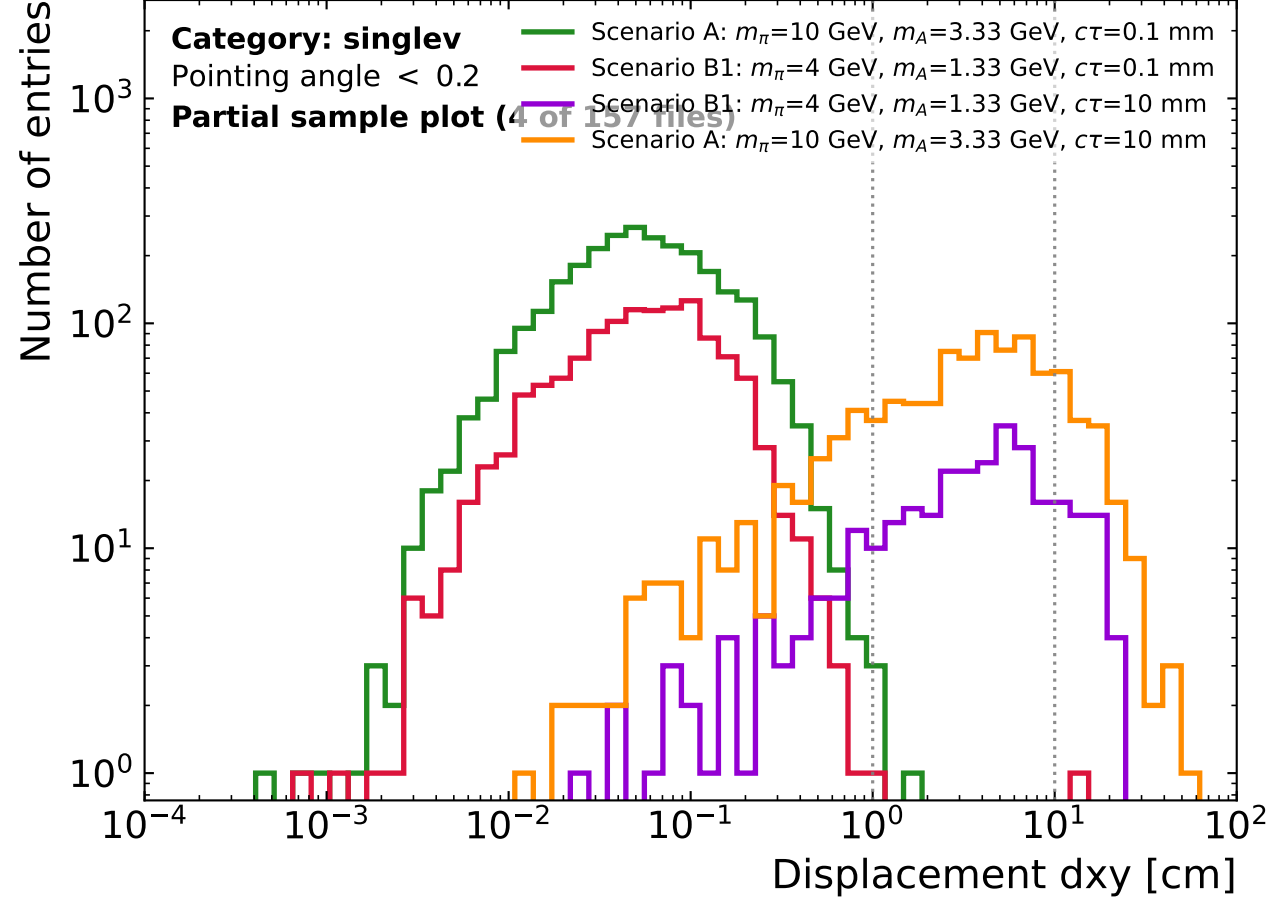
CMS Simulation Preliminary

(13.6 TeV, 2024)



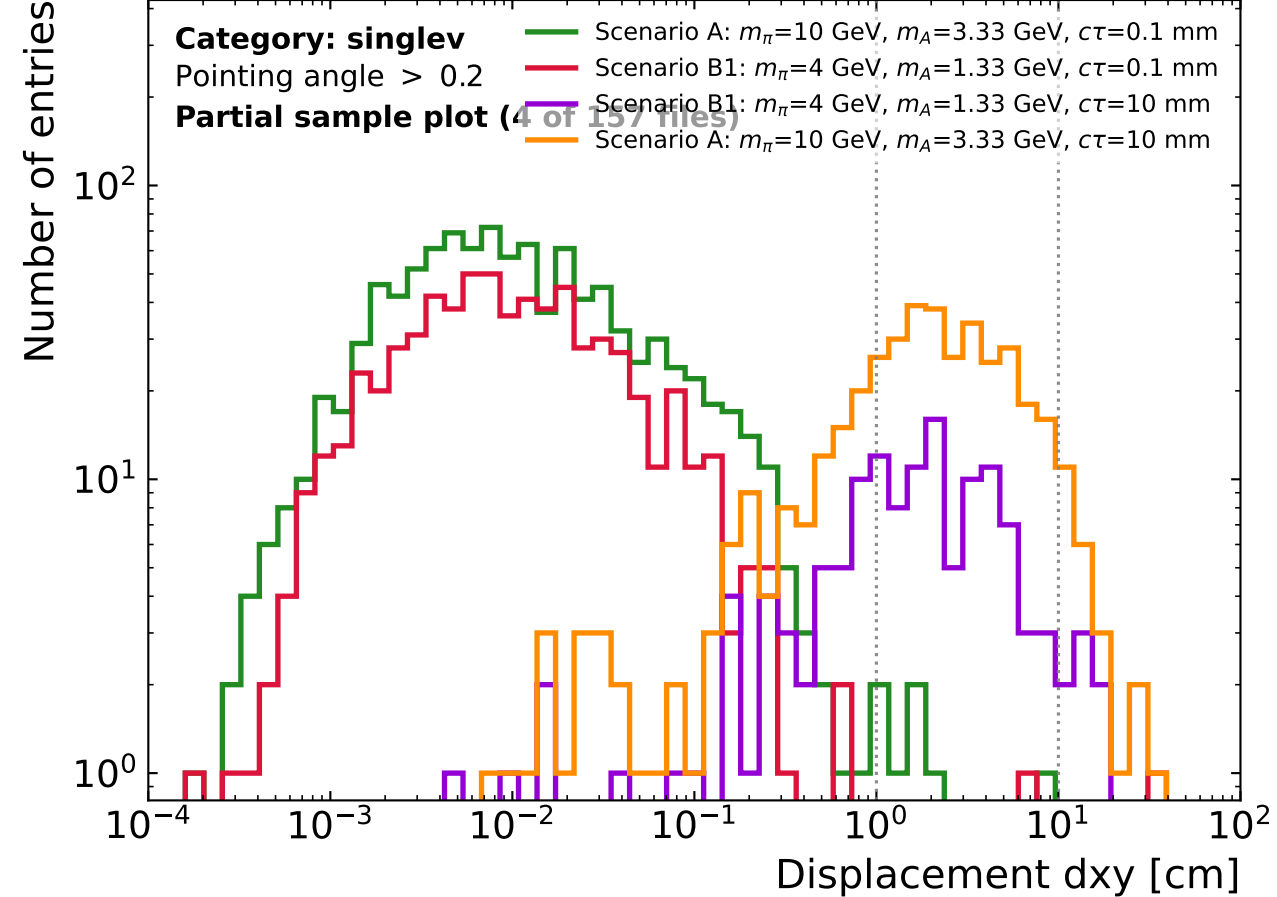
singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary (13.6 TeV, 2024)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$	2,796	2	0	2,798
B1 $c\tau=0.1$	1,260	0	1	1,261
B1 $c\tau=10$	53	200	45	298
A $c\tau=10$	212	628	155	995

CMS Simulation Preliminary (13.6 TeV, 2024)

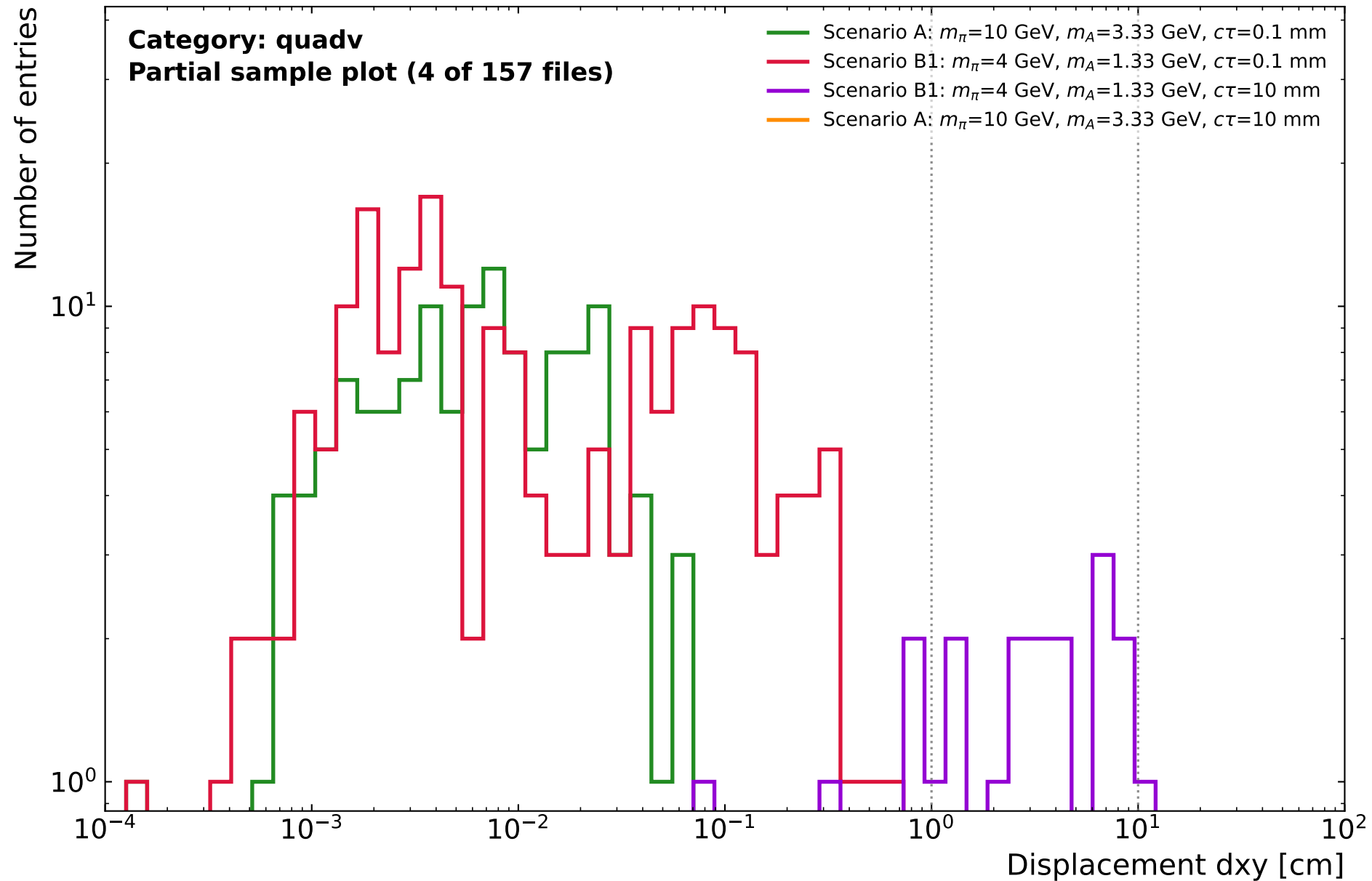


	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$	1,010	6	0	1,016
B1 $c\tau=0.1$	659	1	1	661
B1 $c\tau=10$	44	83	7	134
A $c\tau=10$	115	269	24	408

quadv: displacement of the binning vertex (fourmuonSV_dxy)
dotted: dxy category edges (1, 10 cm)

CMS *Simulation Preliminary*

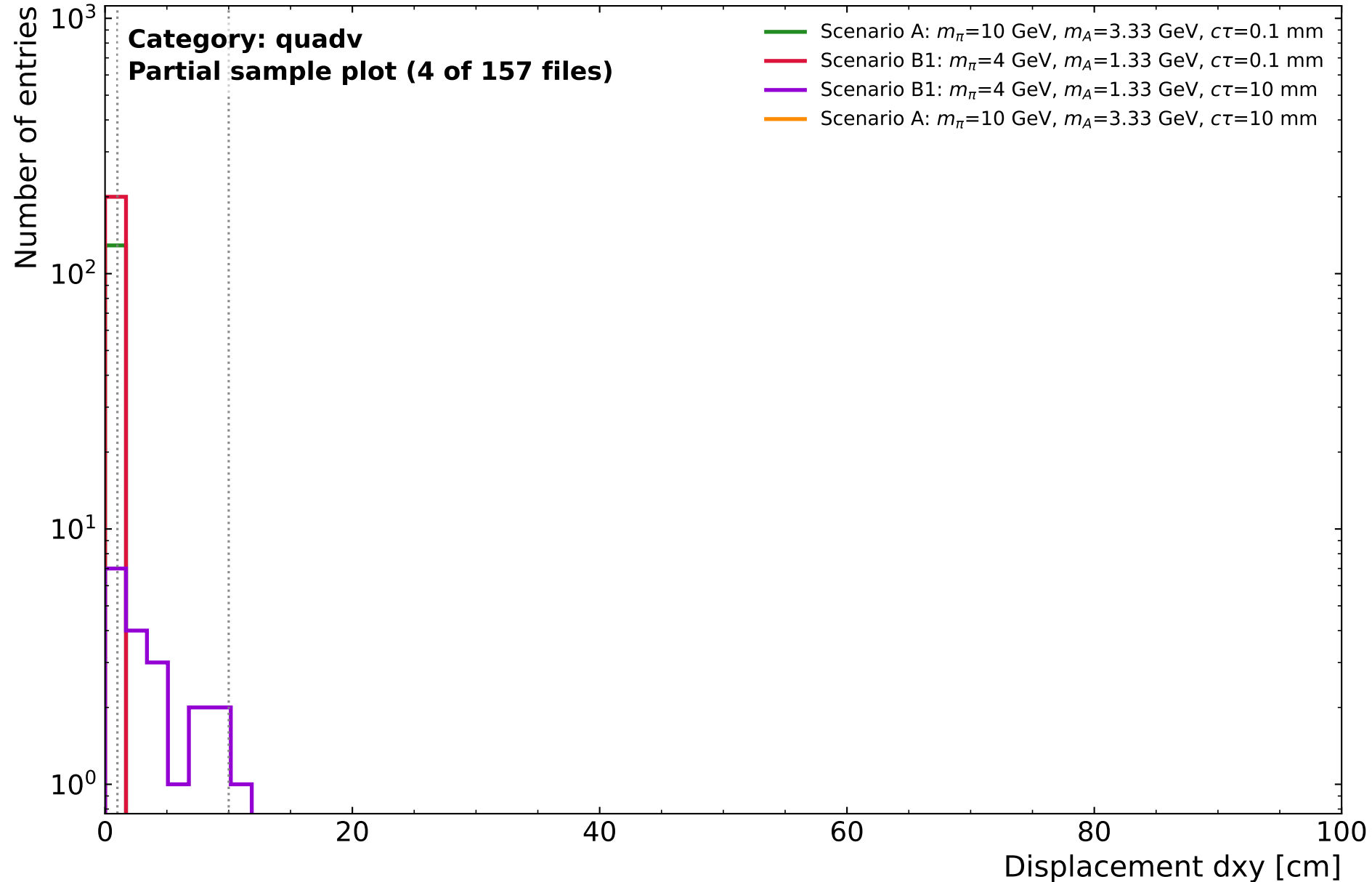
(13.6 TeV, 2024)



quadv: displacement of the binning vertex (fourmuonSV_dxy)
dotted: dxy category edges (1, 10 cm)

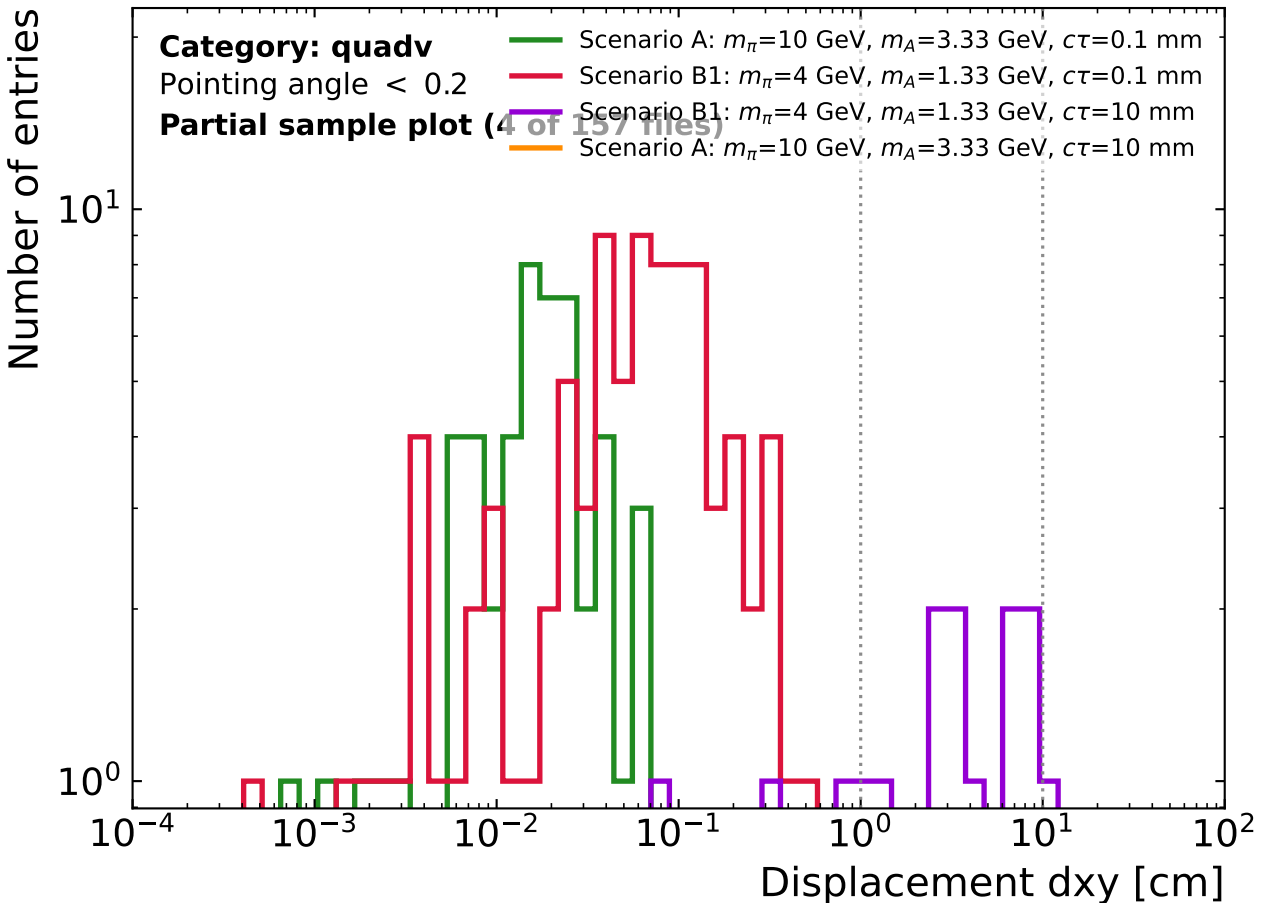
CMS *Simulation Preliminary*

(13.6 TeV, 2024)



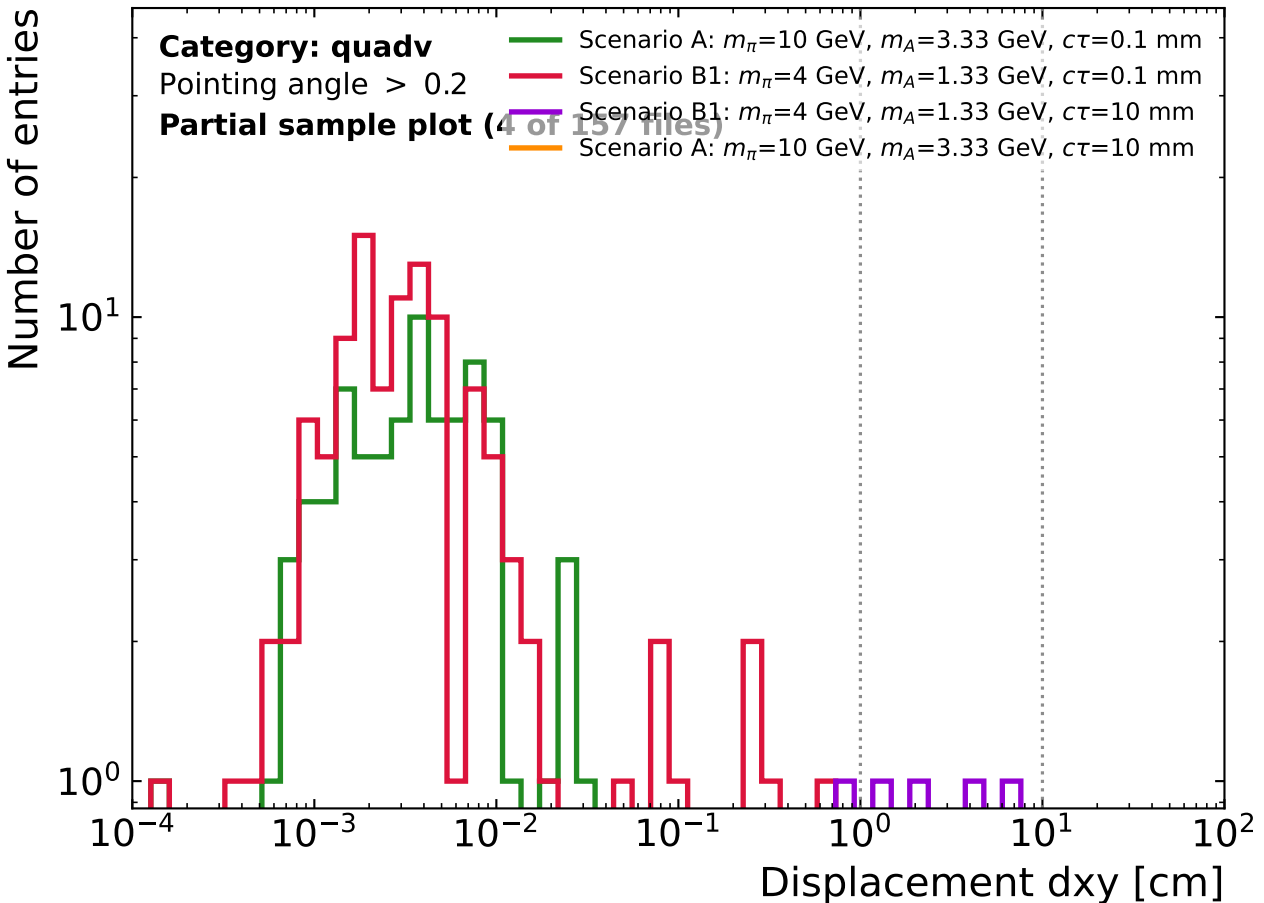
quadv: displacement of the binning vertex (fourmuonSV_dxy), split by pointing angle
dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary (13.6 TeV, 2024)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$	51	0	0	51
B1 $c\tau=0.1$	90	0	0	90
B1 $c\tau=10$	3	11	1	15
A $c\tau=10$	0	0	0	0

CMS Simulation Preliminary (13.6 TeV, 2024)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$	78	0	0	78
B1 $c\tau=0.1$	110	0	0	110
B1 $c\tau=10$	1	4	0	5
A $c\tau=10$	0	0	0	0

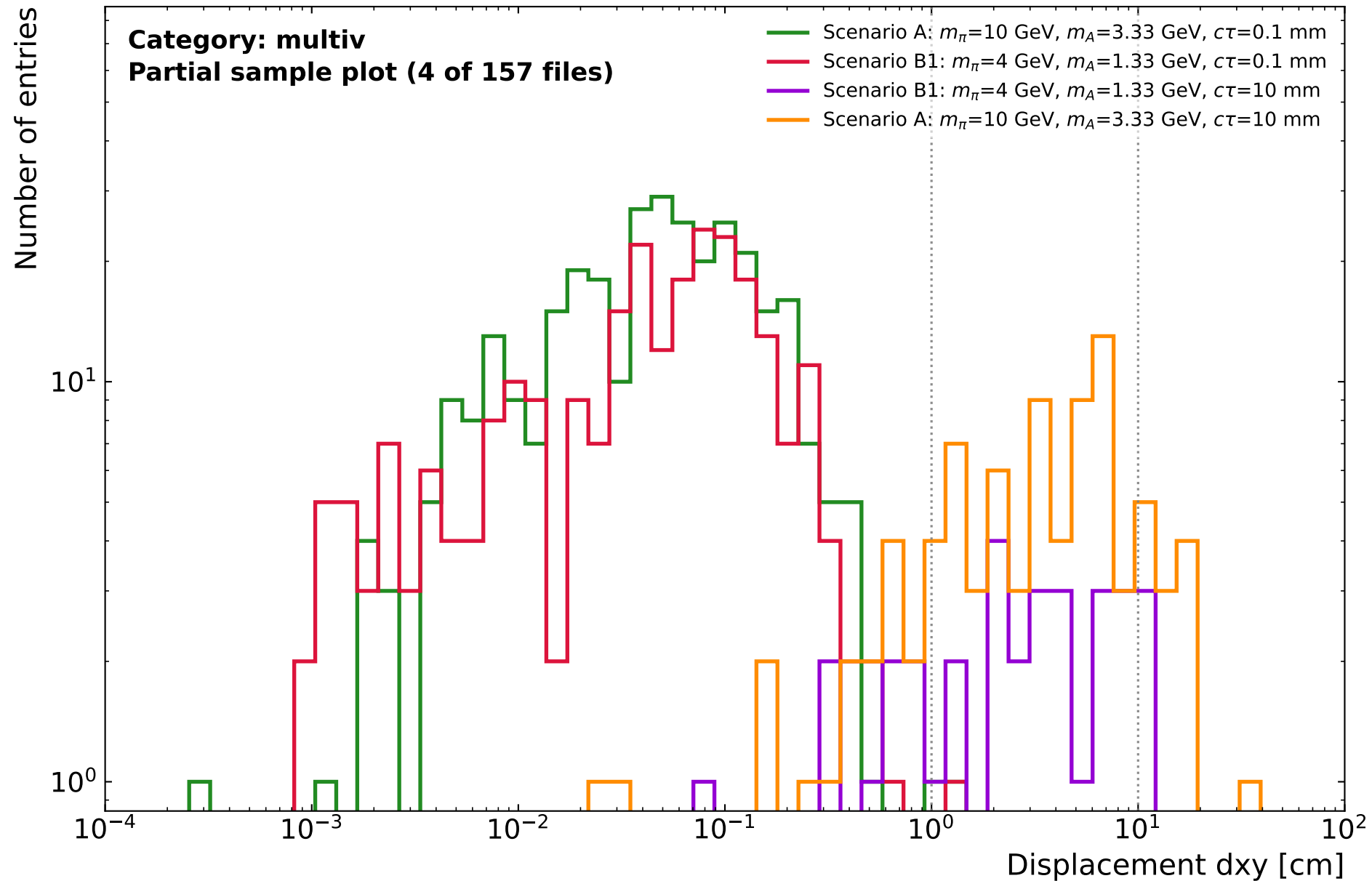
multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
dotted: dxy category edges (1, 10 cm)

CMS *Simulation Preliminary*

(13.6 TeV, 2024)

Category: multiv
Partial sample plot (4 of 157 files)

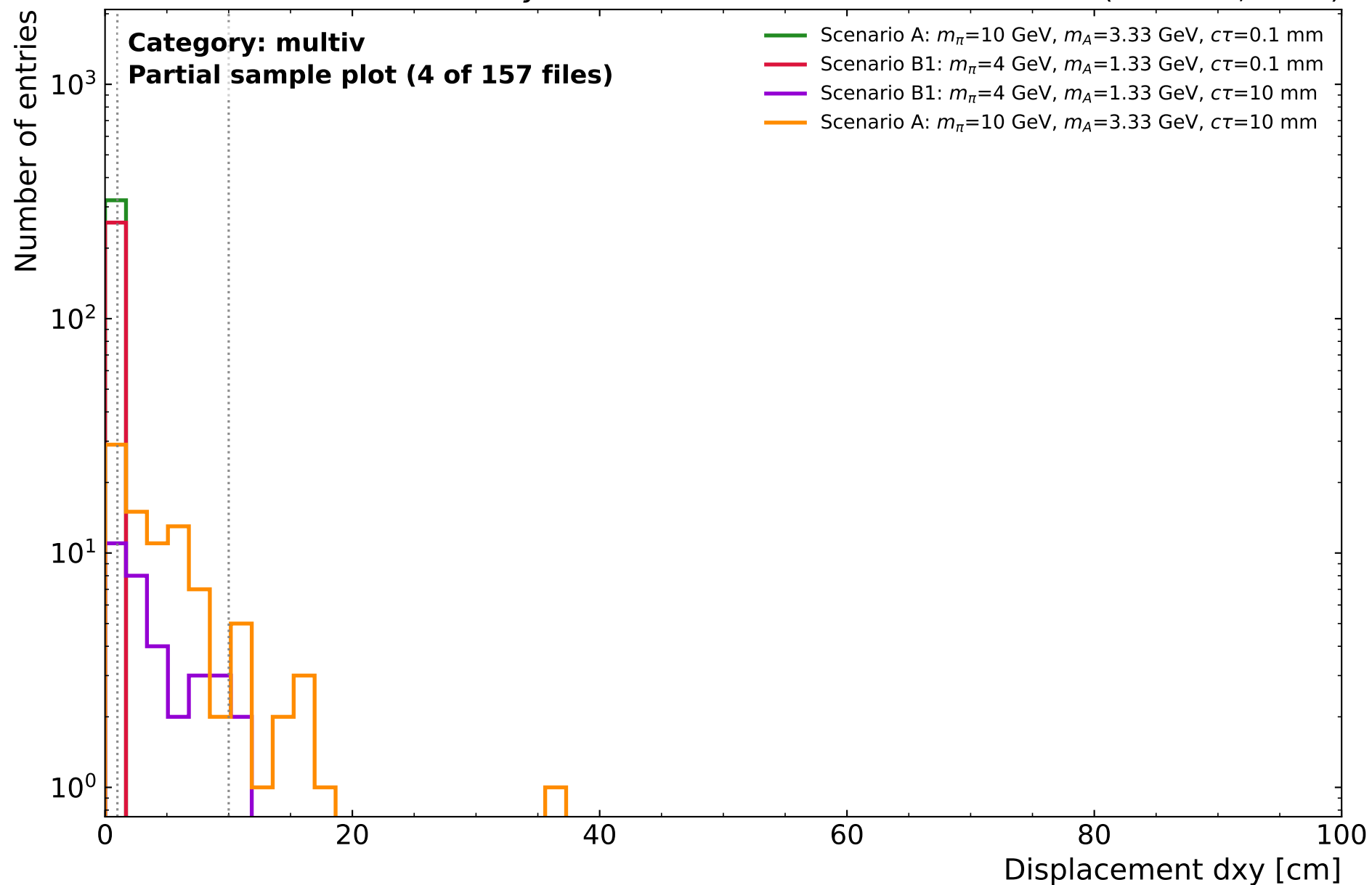
- Scenario A: $m_\pi=10$ GeV, $m_A=3.33$ GeV, $c\tau=0.1$ mm
- Scenario B1: $m_\pi=4$ GeV, $m_A=1.33$ GeV, $c\tau=0.1$ mm
- Scenario B1: $m_\pi=4$ GeV, $m_A=1.33$ GeV, $c\tau=10$ mm
- Scenario A: $m_\pi=10$ GeV, $m_A=3.33$ GeV, $c\tau=10$ mm



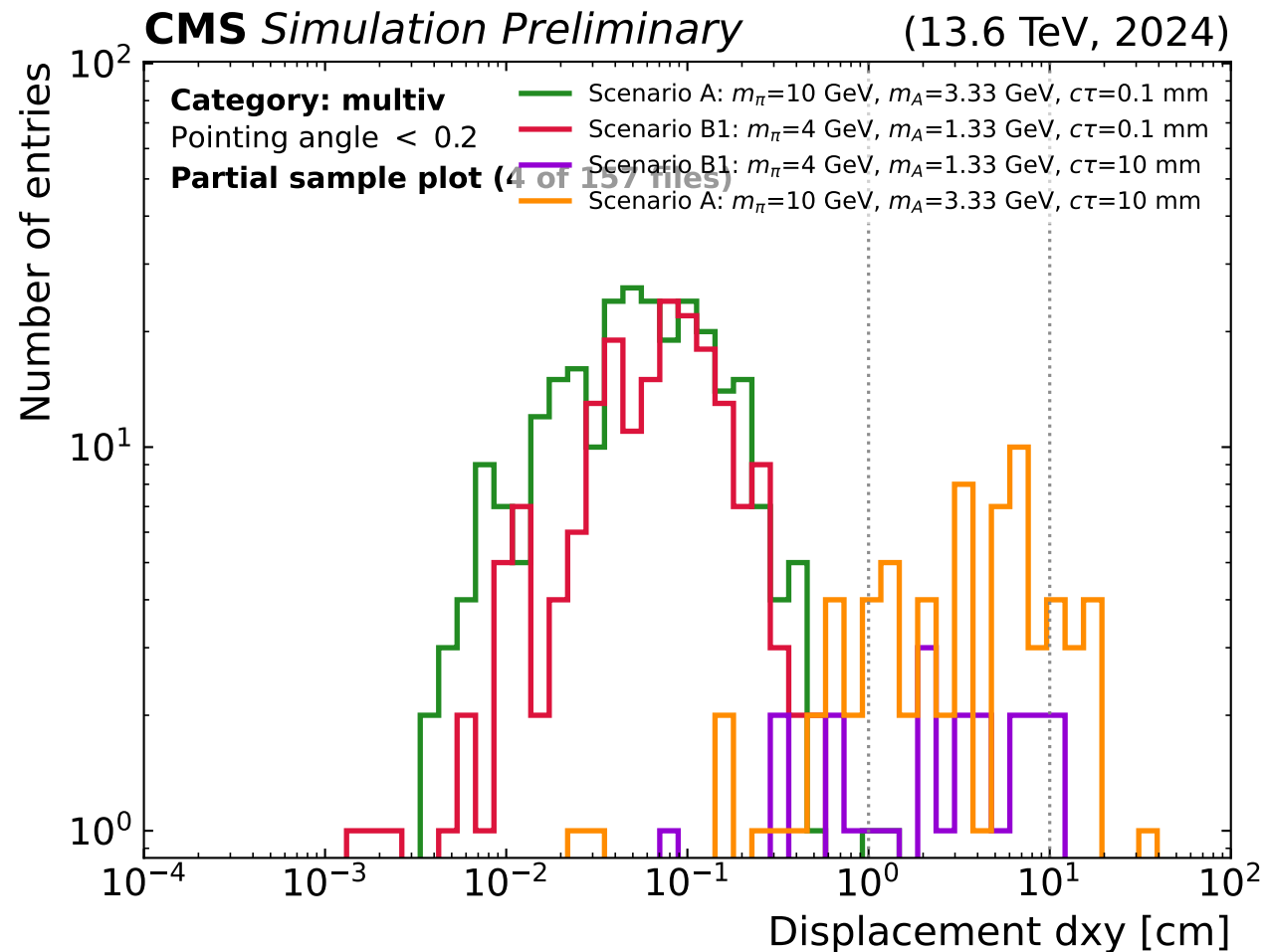
multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
dotted: dxy category edges (1, 10 cm)

CMS Simulation Preliminary

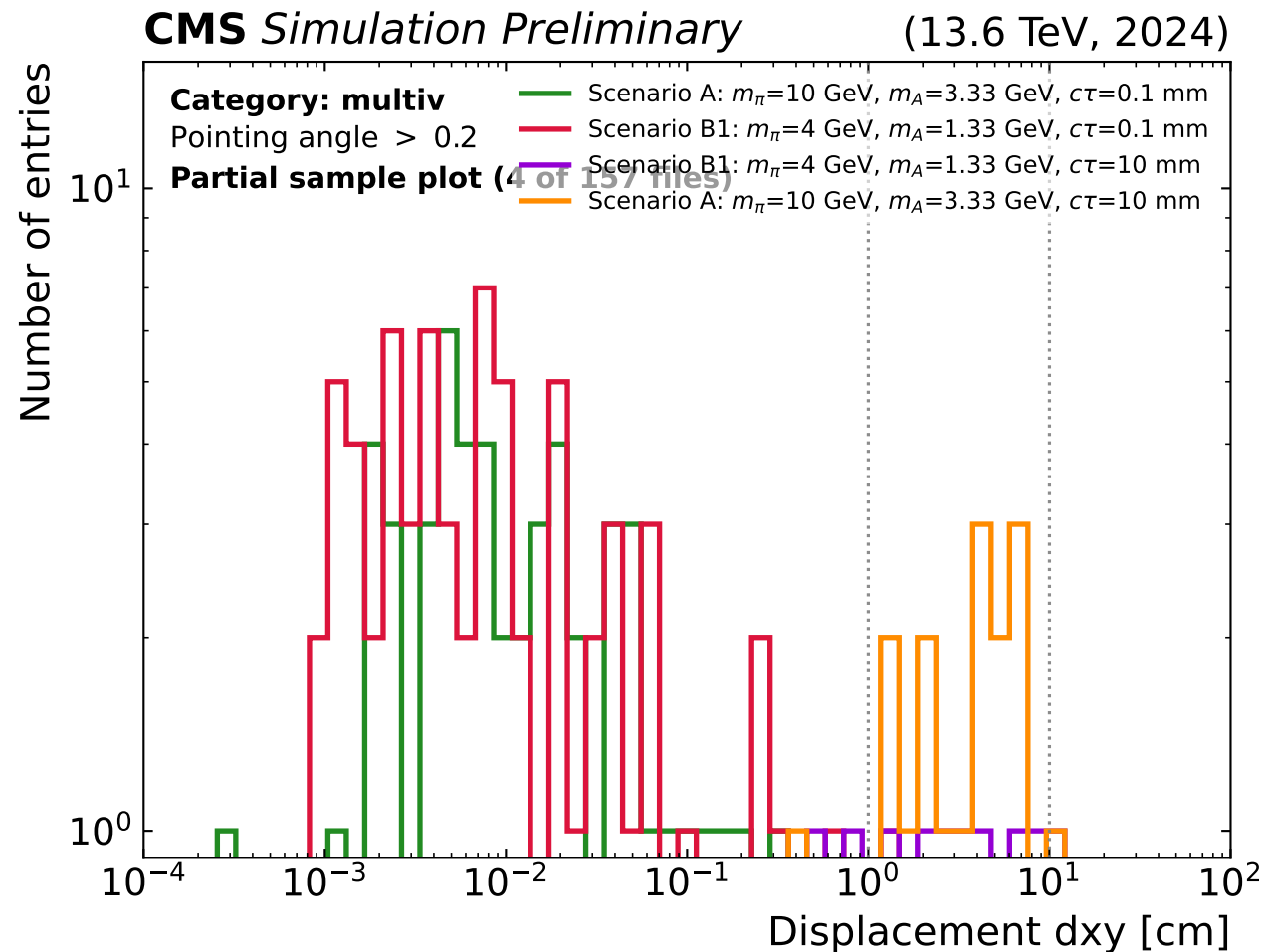
(13.6 TeV, 2024)



multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
dotted: dxy category edges (1, 10 cm)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$	266	2	0	268
B1 $c\tau=0.1$	189	0	0	189
B1 $c\tau=10$	6	15	2	23
A $c\tau=10$	17	44	12	73



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$	52	0	0	52
B1 $c\tau=0.1$	67	1	0	68
B1 $c\tau=10$	2	7	1	10
A $c\tau=10$	1	15	1	17